

NBA Statline Forecasting Conditioned on Roster Availability

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DSCI 4350 - Data Science Capstone





66%

of NBA players suffer at least one injury or illness per season.

Evaluated across a 17 season period by the NIH.

Problem Statement



Simple forecasting methods do not react quickly enough to roster changes.

Sports Betting Companies:

- Frequency of changes makes it difficult to adjust betting lines
- Creates uncertainty and unnecessary exposure concerns
- Points, Rebounds, Assists (PRA) evaluated differently

NBA Team Analytics Departments:

- Addressing team needs when players get injured
- Scouting potential replacements for long-term injuries
- Predictions and information for matchups based on injury situation



Project Goal + Success Criteria

- Injury-aware forecasting system
- Focuses on points, rebounds, assists
- Predicts based on:
 - Roster Availability
 - Player Statistics
 - Matchup Context
- Success Criteria:
 - Lower prediction error than baseline models
 - Evaluate using MAE/RMSE
 - Forecasts respond to injury-driven changes
 - Reproducible ETL and modeling pipeline
 - Lightweight API / Dashboard Output

PLAYER	MIN ▼	USG%	PTS	REB	AST
Nickeil Alexander-Walker	719.0	25.5	21.6	5.0	5.0
Onyeka Okongwu	556.8	19.4	16.9	9.9	3.6
Mouhamed Gueye	529.9	12.8	10.3	9.3	1.6
Luke Kennard	515.5	13.6	15.1	3.7	3.0
Jalen Johnson	443.4	28.9	23.8	12.5	9.1
Vít Krejčí	436.8	16.7	15.7	3.7	2.8
Keaton Wallace	332.4	17.4	10.7	3.8	6.6
Zaccharie Risacher	325.3	18.1	11.6	8.1	2.2
CJ McCollum	322.3	29.1	22.7	4.5	5.4

Infrastructure and Tool Setup

- Communication
 - Slack workspace for project communication and coordination
- Project Management
 - Trello board used to organize Agile sprint tasks and track progress
- Codebase
 - GitHub repository used to store source code, notebooks, documentation
- Documentation
 - Google Drive used to demonstrate project plan and documentation
- Experiment Tracking
 - Weights & Biases used to log experiments and track model performance



Business Understanding

Problem:

- NBA injuries and lineup changes significantly affect player production
- Simple forecasting methods often fail to adjust quickly enough

Stakeholders:

- Sports betting companies
- NBA analytics departments

Primary Objective:

- Predict specific stats (points, rebounds, assists) under injury-driven roster availability scenarios
- Develop a dashboard to display these results for stakeholder use

Success Metrics:

- Baseline comparison, MAE, RMSE



Data Acquisition

NBA Player Statistics:

- Collected from **nba_api**
- Game-level statistics for all players
- Includes major statistics: points, rebounds, assists, minutes, opponent, etc.

NBA Injury Reports:

- Official NBA reports provide player availability information
- Used to determine roster availability and lineup context
- <https://official.nba.com/nba-injury-report-2025-26-season/>

Additional Context:

- Matchup / game information from <https://www.nba.com/stats/players>

Injury Report: 03/13/26 06:45 AM

Game Date	Game Time	Matchup	Team	Player Name	Current Status	Reason
03/13/2026	07:30 (ET)	CLE@DAL	Cleveland Cavaliers	Allen, Jarrett	Out	Injury/Illness - Right Knee; Tendinitis
				Ellis, Keon	Available	Injury/Illness - Left Index Finger; Fracture
				Harden, James	Available	Injury/Illness - Right Thumb; Fracture
				Proctor, Tyrese	Out	Injury/Illness - Right Quadricep; Strain
				Sarr, Olivier	Out	G League - Two-Way
				Strus, Max	Out	Injury/Illness - Left Foot; Surgery - Jones Fracture
				Dallas Mavericks		NOT YET SUBMITTED
		MEM@DET	Memphis Grizzlies			NOT YET SUBMITTED
						Detroit Pistons
		NYK@IND	New York Knicks	Hart, Josh	Questionable	Injury/Illness - Left Knee; Soreness
				Jones, Dillon	Out	G League - Two-Way
				McBride, Miles	Out	Injury/Illness - Pelvic; Core Muscle Surgery
Towns, Karl Anthony	Questionable			Injury/Illness - Bilateral Knee; Bilateral Knee Soreness		

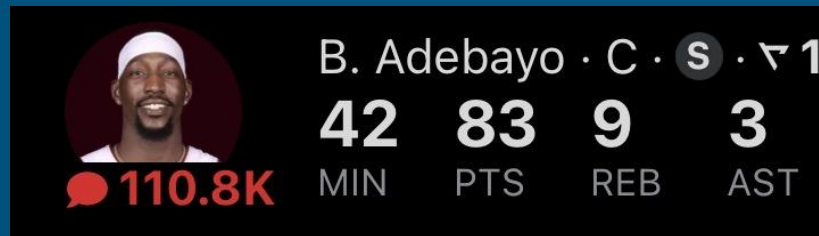
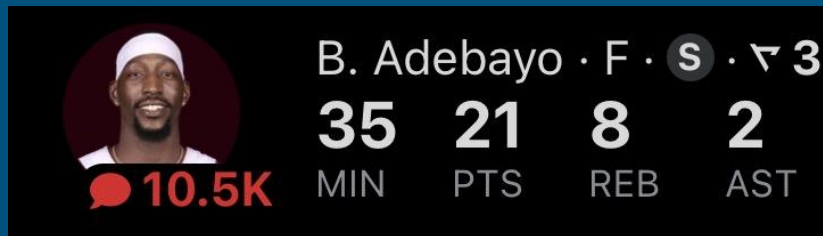
Target Variable

Predict player game performance by three essential offensive metrics: PRA

- Points, Rebounds, Assists

Prediction task: supervised regression modeling

- Player statistics, roster availability, matchup context



Exploratory Data Analysis

Data Quality and Preparation.

Dataset Source: nba_api (player game statistics)

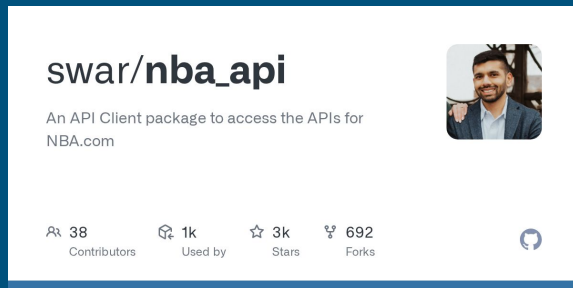
Variables Analyzed: points, rebounds, assists, minutes

Data Quality Checks:

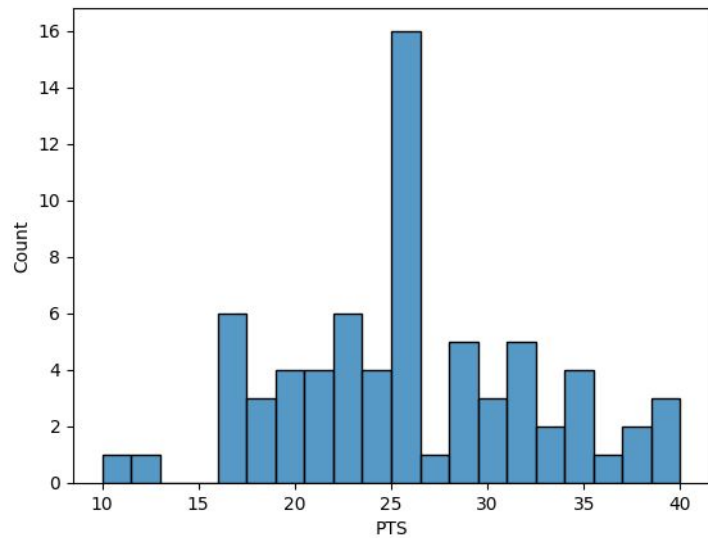
- Verified dataset completeness ✓
- Checked dataset for missing values ✓
- Confirmed numeric data types for analysis ✓

Data Preparation:

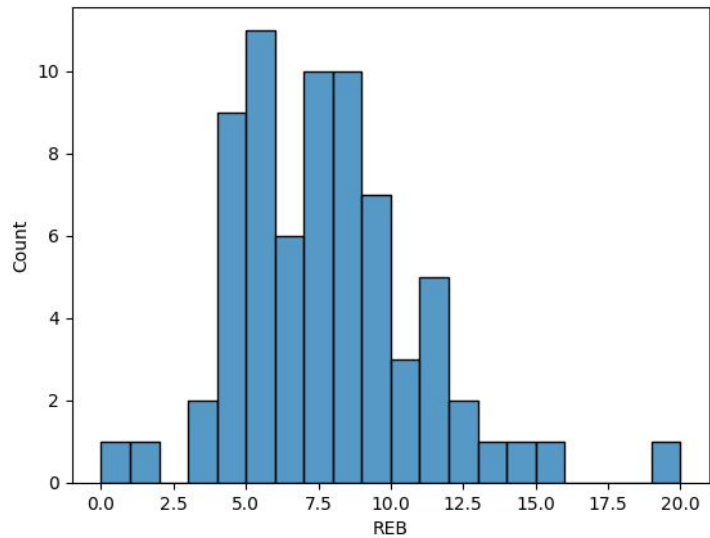
- Selected key performance metrics and relevant columns
- Structured dataset for EDA



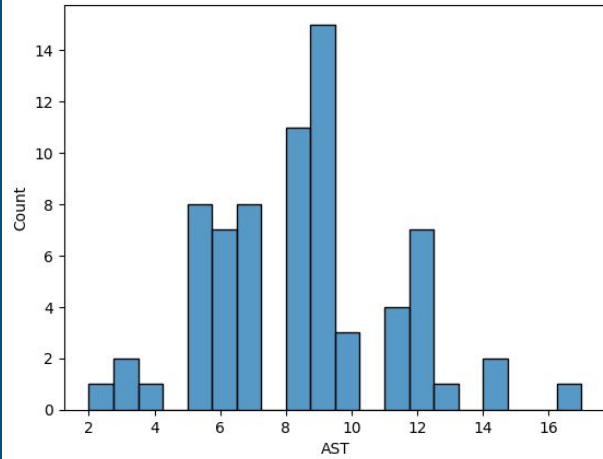
Distribution of Points

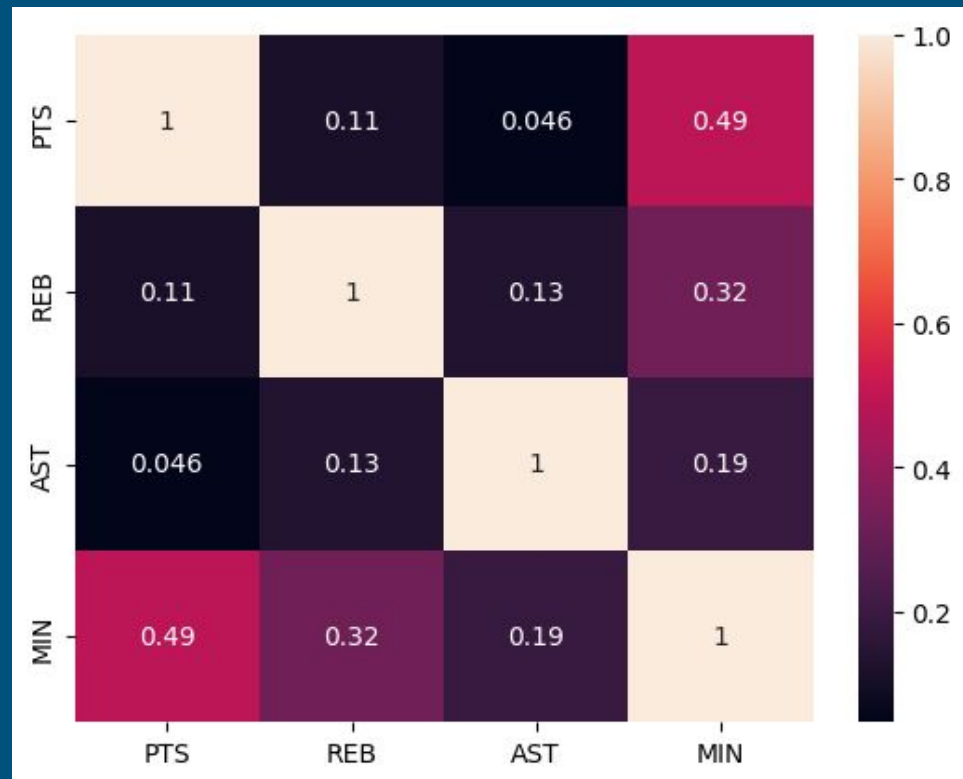
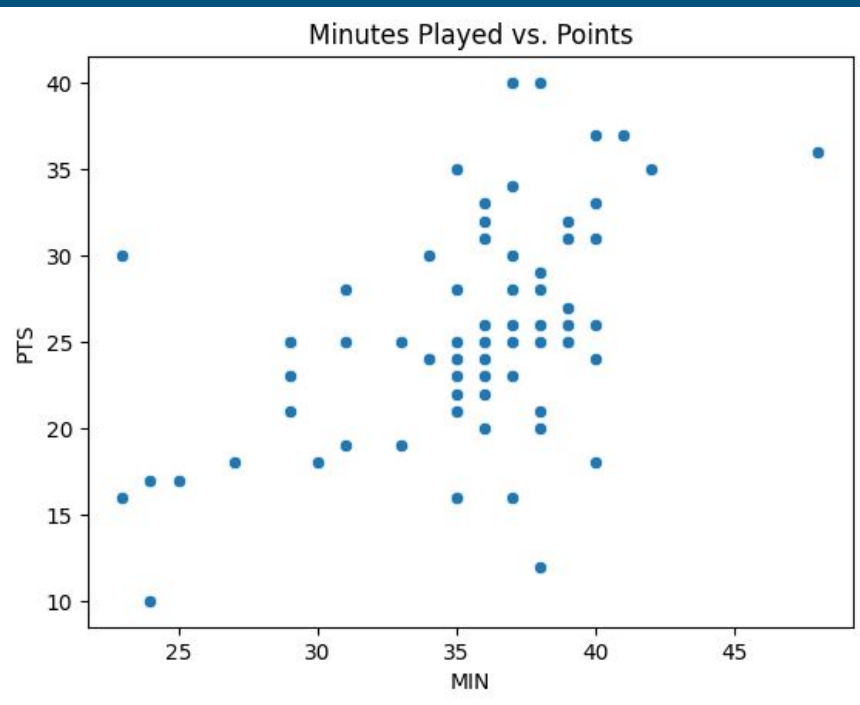


Distribution of Rebounds



Distribution of Assists





Initial Findings

- Correlation between minutes played and points (0.49)
 - Rebounds and Assists show weaker correlation with minutes
- Confirmation: performance distributions = typical production ranges
 - 20-30 points, 5-10 rebounds, 6-10 assists
- Insignificant correlation between the three target variables (PRA)
- Main EDA Takeaway: Minutes played appears to be the strongest structural driver of correlation.
- Further motivates the need to incorporate roster availability and minutes projections into the forecasting model.

Summary



Problem: NBA roster changes significantly impact player performance and usage.

Progress:

- Established project infrastructure
- Defined forecasting target variables (PRA)
- Conducted EDA using nba_api player statistics

Key Insight: Player opportunity, particularly minutes played, appears to strongly influence statistical scoring output

Next Steps:

- Expand dataset across multiple players and seasons
- Develop predictive models incorporating roster availability / parsed injury reports
- Evaluate model error and accuracy using MAE / RMSE